SOC Fundamentals and Operations Report

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1. Introduction

This report provides an overview of SOC operations, tools, workflows, and practical implementation in a simulated lab environment.

2. SOC Fundamentals

Purpose: Threat detection, incident response, continuous monitoring.

Roles: Tier 1, Tier 2, Tier 3 analysts, SOC Manager.

Functions: Log analysis, alert triage, threat intelligence.

3. Security Monitoring

Objectives: Detect anomalies, unauthorized access, policy violations.

Tools: Wazuh, Elastic SIEM, Wireshark.

Metrics: False positives, MTTD.

4. Log Management

Lifecycle: Collection → Normalization → Storage → Analysis.

Practical: Used Fluentd and Logstash for log collection and JSON conversion.

5. Security Tools

SIEM: Wazuh | IDS: Snort | Scanner: Nessus | Endpoint: Osquery.

6. Incident Response

Lifecycle: Preparation → Identification → Containment → Eradication → Recovery.

7. Practical Work

Performed log analysis using Event ID 4625 (failed login).

Simulated brute-force attack and detected suspicious activity.

Configured alert rules and dashboards.

8. Documentation

Date/Time	Source IP	Event ID	Description	Action
2026-03-18	192.168.1.10	4625	Failed logins	Blocked IP

9. Conclusion

This project demonstrated SOC monitoring, detection, and response using real tools.